Project One: Homelessness

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This file contains the workflow I used for the project topic Homelessness

I choose to work with homeless students in New York state. I picked up this topic because this is probably the social class most affected by homelessness, given that they are young and must make decisions that will guide and impact adult life. Everything I used here is also available on my *GitHub*.

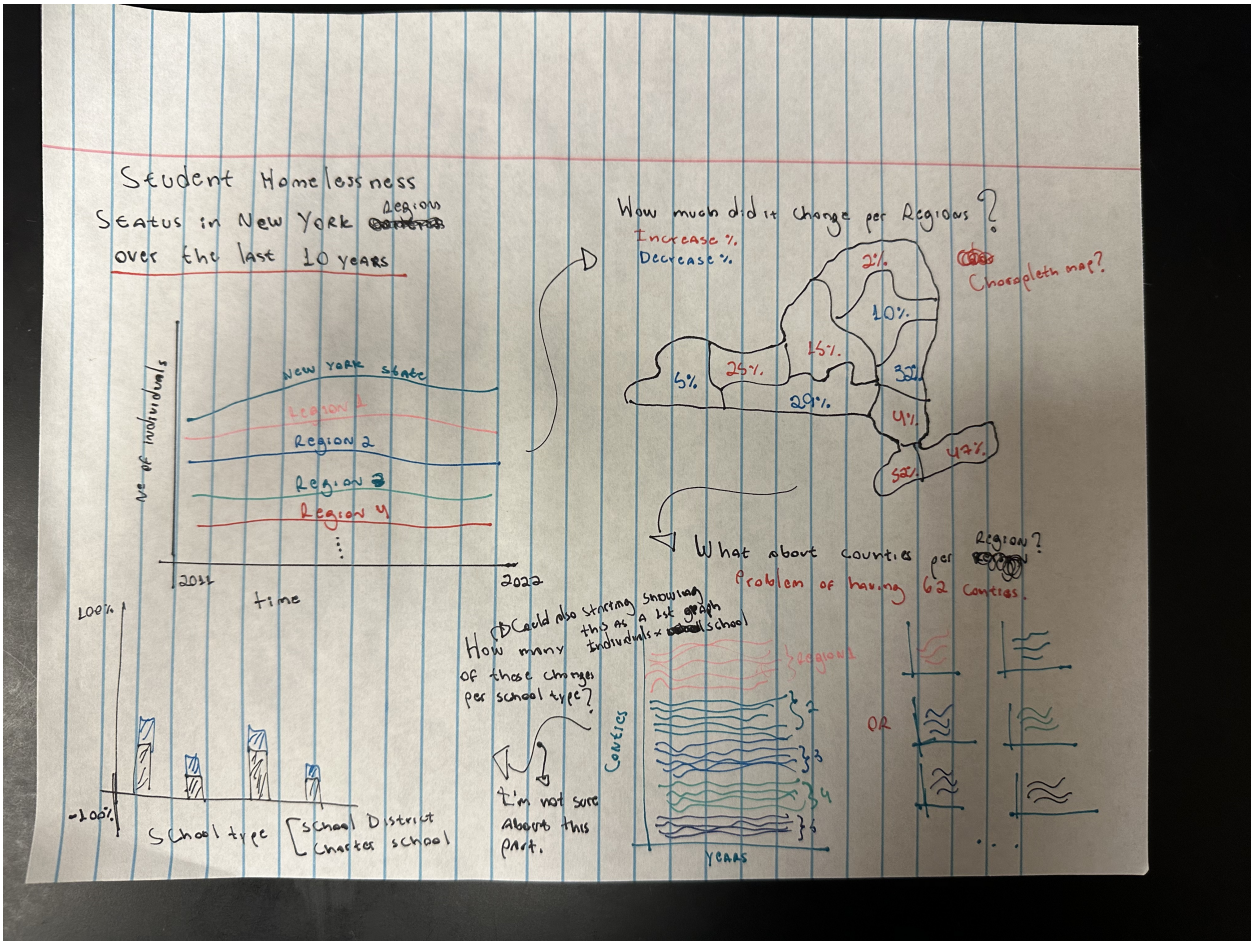
The infochartic was designed mainly in four sections. **1.** The first section includes a chart containing the total number of homeless students from 2009 to 2021. I decided to use a line chart because it is easier to see trends in time. Aligned with this chart, I included a map of students' homeless per county in New York state. I decided to use counties because I want to show that this problem is happening across the whole state.

2. Given that New York has 62 counties and it would be hard to make a chart representing each of them well, I tried to show these changes in the second section using the New York economic region's boundaries. I chose a stream chart to display the data over time because I wanted to show trends in the increased proportion between the areas simultaneously. Besides, it allowed me to avoid lines overlapping within the chart. I included a map showing the proportion increase in 2021 compared to 2009 per economic region. My idea of plotting both chart and map is to show that the rise in homeless students is unrelated to the region area size and that only looking at the percentage increase does not represent the number of students per region well.

3. In the third section, I use a line chart to show which schools these students attend and how disproportional it is over time. I also included a bar chart showing the proportion changes in 2021 compared to 2009.

4. The fourth section is dedicated to the show the students' proportion per school grade over time, using a bumpchart. My choice for a bumpchat was because it allowed me to show the overlap between lines and trends better than the line chart. This section also has a bar chart comparing 2021 to 2009 for each grade. Since I do not have the data relative to students' age, I considered using grades to proxy how old these people living in homeless conditions are.

The figure below is a free-hand draft I made before designing the infographic using Adobe Illustrator®.


Student Homelessness
Seatus in New York Region
over the last 10 years
No. of individuals
New York state
Region 1
Region 2
Region 3
Region 4
2032
2022
time
How much did it change per Regions?
Increase %
Decrease %
2%
10%
15%
5%
25%
20%
30%
40%
50%
47%
Choropleth map?
What about counties per Region?
Problem of having 62 Counties.
100%
-100%
School type
School District
Charter school
How many of those changes per school type?
Could also starting showing this as a bar graph
individuals x school
I'm not sure About this part.
Counties
Region 1
1
3
4
6
Years
OR

Loading the packages

```
if(!require(tidyverse))install.packages("tidyverse", dependencies = TRUE)
if(!require(readxl)) install.packages("readxl", dependencies = TRUE)
if(!require(purrr)) install.packages("purrr", dependencies = TRUE)
if(!require(writexl)) install.packages("writexl", dependencies = TRUE)
if(!require(openxlsx)) install.packages("openxlsx", dependencies = TRUE)
if(!require(data.table)) install.packages("data.table", dependencies = TRUE)
```

Setting directories and reading tables

```
path <- "C:/Users/Yuri/My Drive/PhD/00_UM/03_graduate/00_courses/02_2nd_semester/02_infographic_I/01_cla
path

regions <- readr::read_csv("C:/Users/Yuri/My Drive/PhD/00_UM/03_graduate/00_courses/02_2nd_semester/02_
  unique()
regions

file_path <- list.files(path, pattern="\\.xlsx$", full.names = TRUE)
file_path


#For each file in file_path
homeless.tables <- purrr::map(file_path, ~
                          #For each sheet
                          purrr::map(2:3, function(i) {
                            #Read the file with particular sheet nummber
                            openxlsx::read.xlsx(.x, sheet=i, startRow=1)}) %>%
                          purrr::reduce(dplyr::full_join) %>%
                          #Remove all NA rows
                          #dplyr::filter(Reduce(`|`, across(.fns = ~!is.na(.)))) %>%
                          #Add Year column at 1st position
                          dplyr::mutate(YEAR = tools::file_path_sans_ext(basename(.x)), .before = 1))

homeless.tables
```

Binding the tables in a single one

```
homeless.tables.bind <- data.table::rbindlist(homeless.tables,
                                              use.names=TRUE,
                                              fill=FALSE,
                                              idcol=TRUE) %>%
  dplyr::rename_all(., .funs = toupper) %>%
  dplyr::filter(!"TOTAL.HOMELESS" == "s") %>%
  dplyr::mutate(TOTAL.HOMELESS = as.numeric(TOTAL.HOMELESS),
                YEAR = as.numeric(stringr::str_extract(YEAR, ".0..")))) %>%
  dplyr::left_join(regions) %>%
  stats::na.omit() %>%
  dplyr::select(c(3,4,6,5,2))
```

```
## Warning in mask$eval_all_mutate(quo): NAs introduced by coercion
```

```
homeless.tables.bind

#write_csv(homeless.tables.bind, "C:/Users/Yuri/My Drive/PhD/00_UM/03_graduate/00_courses/02_2nd_semest
```

Summarizing the tables by total individuals per year

```
homeless.tables.bind.year <- homeless.tables.bind %>%
  dplyr::group_by(YEAR) %>%
  dplyr::summarise(TOTAL.HOMELESS = sum(TOTAL.HOMELESS))
homeless.tables.bind.year

#write_csv(homeless.tables.bind.year, "C:/Users/Yuri/My Drive/PhD/00_UM/03_graduate/00_courses/02_2nd_s

homeless.tables.bind.year.diff <- homeless.tables.bind.year %>%
  dplyr::filter(YEAR == 2009 | YEAR ==  2021) %>%
  tidyr::pivot_wider(names_from = YEAR, values_from = TOTAL.HOMELESS) %>%
  dplyr::mutate(DIFF = `2021`-`2009`,
                PCT_INCREASE = ((DIFF*100)/(`2009`))) %>%
  dplyr::filter(PCT_INCREASE != 0) %>%
  dplyr::filter_all(all_vars(!is.infinite(.)))
homeless.tables.bind.year.diff

#write_csv(homeless.tables.bind.year.diff, "C:/Users/Yuri/My Drive/PhD/00_UM/03_graduate/00_courses/02_
```

Summarizing the tables by total individuals per year and region

```
homeless.tables.bind.year.region <- homeless.tables.bind %>%
  dplyr::group_by(REGION, YEAR) %>%
  dplyr::summarise(TOTAL.HOMELESS = sum(TOTAL.HOMELESS)) %>%
  dplyr::ungroup()
homeless.tables.bind.year.region

#write_csv(homeless.tables.bind.year.region, "C:/Users/Yuri/My Drive/PhD/00_UM/03_graduate/00_courses/0

homeless.tables.bind.year.region.diff <- homeless.tables.bind.year.region %>%
  dplyr::filter(YEAR == 2009 | YEAR ==  2021) %>%
  tidyr::pivot_wider(names_from = YEAR, values_from = TOTAL.HOMELESS) %>%
  dplyr::mutate(DIFF = `2021`-`2009`,
                PCT_INCREASE = ((DIFF*100)/(`2009`))) %>%
  dplyr::filter(PCT_INCREASE != 0) %>%
  dplyr::filter_all(all_vars(!is.infinite(.))) %>%
  tidyr::pivot_longer(cols = 2:3, values_to = "INDI", names_to = "YEAR") %>%
  dplyr::select(c(1,4,5,2,3))

homeless.tables.bind.year.region.diff

#write_csv(homeless.tables.bind.year.region.diff, "C:/Users/Yuri/My Drive/PhD/00_UM/03_graduate/00_cour
```

```
homeless.tables.bind.year.region.pct.diff <- homeless.tables.bind.year.region %>%
  #dplyr::filter(YEAR == 2009 | YEAR ==  2021) %>%
  tidyr::pivot_wider(names_from = YEAR, values_from = TOTAL.HOMELESS) %>%
  dplyr::mutate(`2011i` = (((`2009`-`2011`)*100)/(`2009`)),
                `2012i` = (((`2011`-`2012`)*100)/(`2011`)),
                `2013i` = (((`2012`-`2013`)*100)/(`2012`)),
                `2014i` = (((`2013`-`2014`)*100)/(`2013`)),
                `2015i` = (((`2014`-`2015`)*100)/(`2014`)),
                `2016i` = (((`2015`-`2016`)*100)/(`2015`)),
                `2017i` = (((`2016`-`2017`)*100)/(`2016`)),
                `2018i` = (((`2017`-`2018`)*100)/(`2017`)),
                `2019i` = (((`2018`-`2019`)*100)/(`2018`)),
                `2020i` = (((`2019`-`2020`)*100)/(`2019`)),
                `2021i` = (((`2020`-`2021`)*100)/(`2020`))) %>%
  dplyr::filter_all(all_vars(!is.infinite(.))) #%>%
  #tidyr::pivot_longer(cols = 2:13, values_to = "DIFF_PCT", names_to = "YEAR") %>%
  #dplyr::select(c(1,3,2)) %>%
  #mutate(DIFF_PCT = round(DIFF_PCT, digits = 0),
  #       DIFF_STS = ifelse(DIFF_PCT >= 0, "Positive", "Negative"))


homeless.tables.bind.year.region.pct.diff

#write_csv(homeless.tables.bind.year.region.pct.diff, "C:/Users/Yuri/My Drive/PhD/00_UM/03_graduate/00_
```

Summarizing the tables by total individuals per year and county

```
homeless.tables.bind.year.county <- homeless.tables.bind %>%
  dplyr::group_by(COUNTY, YEAR) %>%
  dplyr::summarise(TOTAL.HOMELESS = sum(TOTAL.HOMELESS))
homeless.tables.bind.year.county

#write_csv(homeless.tables.bind.year.county, "C:/Users/Yuri/My Drive/PhD/00_UM/03_graduate/00_courses/0

homeless.tables.bind.year.county.diff <- homeless.tables.bind.year.county %>%
  dplyr::filter(YEAR == 2009 | YEAR ==  2021) %>%
  tidyr::pivot_wider(names_from = YEAR, values_from = TOTAL.HOMELESS) %>%
  dplyr::mutate(DIFF = `2021`-`2009`,
                PCT_INCREASE = ((DIFF*100)/(`2009`))) %>%
  dplyr::filter(PCT_INCREASE != 0) %>%
  dplyr::filter_all(all_vars(!is.infinite(.))) %>%
  tidyr::pivot_longer(cols = 2:3, values_to = "INDI", names_to = "YEAR") %>%
  dplyr::select(c(1,4,5,2,3))
homeless.tables.bind.year.county.diff

#write_csv(homeless.tables.bind.year.county.diff, "C:/Users/Yuri/My Drive/PhD/00_UM/03_graduate/00_cour
```

Summarizing the tables by total individuals per year and school type

```
homeless.tables.bind.year.school <- homeless.tables.bind %>%
  dplyr::group_by(LEA.TYPE, YEAR) %>%
  dplyr::summarise(TOTAL.HOMELESS = sum(TOTAL.HOMELESS))
homeless.tables.bind.year.school

#write_csv(homeless.tables.bind.year.school, "C:/Users/Yuri/My Drive/PhD/00_UM/03_graduate/00_courses/0

homeless.tables.bind.year.school.diff <- homeless.tables.bind.year.school %>%
  dplyr::filter(YEAR == 2009 | YEAR ==  2021) %>%
  tidyr::pivot_wider(names_from = YEAR, values_from = TOTAL.HOMELESS) %>%
  dplyr::mutate(DIFF = `2021`-`2009`,
                PCT_INCREASE = ((DIFF*100)/(`2009`))) %>%
  dplyr::filter(PCT_INCREASE != 0) %>%
  dplyr::filter_all(all_vars(!is.infinite(.))) %>%
  tidyr::pivot_longer(cols = 2:3, values_to = "INDI", names_to = "YEAR") %>%
  dplyr::select(c(1,4,5,2,3))
homeless.tables.bind.year.school.diff

#write_csv(homeless.tables.bind.year.school.diff, "C:/Users/Yuri/My Drive/PhD/00_UM/03_graduate/00_cour
```

Summarizing the tables by grades

```
homeless.ny_regions <- readr::read_csv("C:/Users/Yuri/My Drive/PhD/00_UM/03_graduate/00_courses/02_2nd_
  unique()

homeless.ny_grades_regions <- readr::read_csv("C:/Users/Yuri/My Drive/PhD/00_UM/03_graduate/00_courses/
  dplyr::left_join(homeless.ny_regions) %>%
  dplyr::select(c(20, 3:14,19)) %>%
  tidyr::pivot_longer(cols = 2:13, values_to = "Grades", names_to = "Age") %>%
  #readr::write_csv("00_data/ny_homeless_student_grades_region.csv") %>%
  dplyr::group_by(Year, Age) %>%
  summarise(value = sum(Grades)) %>%
  dplyr::ungroup() %>%
  dplyr::filter(Year == "2009" | Year == "2021") %>%
  tidyr::pivot_wider(names_from = "Year", values_from = "value") %>%
  dplyr::mutate(DIFF = `2021`-`2009`,
                PCT_INCREASE = ((DIFF*100)/(`2009`))) %>%
  dplyr::filter(PCT_INCREASE != 0) #%>%
  #readr::write_csv("00_data/ny_homeless_student_grades_region_diff.csv")
```

First Version

The page below is the first version of the project proposed here. Some considerations I have for the second version:

1. Maybe remove the bar chart of section three or include it within the line chart in this section since this bar chart seems a little redundant.

2. Finding some neat way to replace or show the bumpchat chart in section four.

3. Draw an image at the top of the last chart of section four.

4. Include information about where these students live when they are not in school.